

Customer and Sales Performance Analysis

A SQL Data Analysis Project Using the Northwind Database

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Project Overview

This project analyzes a company's customer, sales, employee, product, and shipping data using SQL. The goal was to answer six real business questions that a company owner or manager would genuinely want answered, then translate the findings into practical recommendations.

The dataset used is Northwind, a well-known sample database representing a small business that imports and sells specialty food and beverage products. All queries were written and run using DB Browser for SQLite.

Tools used: SQL (SQLite), DB Browser for SQLite.

Data Quality Checks

Before trusting any result, it's important to check the data itself for issues. Two data quality checks were carried out during this project, and both are included here to show the kind of verification a careful analyst does before reporting numbers to a client.

1. Fake customer records skewing customer spend

While calculating the Top 10 customers by spend, one result stood out: a customer named "IT" appeared at the top with an unusually high total. Investigating further revealed two separate customer records, both named "IT", with every other field (address, city, region, country, phone) blank. These were clearly placeholder or test records rather than real customers, and grouping by name had merged them into one inflated total.

Resolution: both records were excluded from the Top 10 customers analysis, producing a clean, trustworthy result.

2. Best seller ranking depends on the metric used

An early version of the "best-selling products" query ranked products by total revenue. This showed expensive products like Côte de Blaye at the top, while quantity sold was actually very similar across almost every product, from best sellers to worst sellers. Ranking by revenue alone would have overstated how much more popular expensive products really were.

Resolution: the final analysis ranks best and worst sellers by quantity sold (units actually purchased), with revenue reported alongside as supporting context. This gives a fairer picture of what customers actually prefer.

Question 1: Who Are Our Top 10 Customers by Spend?

Why this matters: Knowing who your biggest spenders are helps you protect those relationships with better service or loyalty offers, because losing a top customer often costs more than losing ten small ones.

Rank	Company	Total Spent
1	B's Beverages	6,154,115.34
2	Hungry Coyote Import Store	5,698,023.67
3	Rancho grande	5,559,110.08
4	Gourmet Lanchonetes	5,552,309.81
5	Ana Trujillo Emparedados y helados	5,534,356.65
6	Ricardo Adocicados	5,524,517.31
7	Folies gourmandes	5,505,502.85

B's Beverages is the top customer by total spend, after excluding the two invalid "IT" placeholder records identified during data quality checks.

Question 2: Which Employee Has Made the Most Sales?

Why this matters: This shows management who's driving revenue, who deserves recognition or a bonus, and who might benefit from training or support to improve.

Rank	Employee	Total Sales
1	Margaret Peacock	51,488,395.20
2	Steven Buchanan	51,386,459.10
3	Janet Leverling	50,445,573.76
4	Nancy Davolio	49,659,423.23
5	Robert King	49,651,899.31

Margaret Peacock is the top-performing employee by total sales, though the results are fairly close across the team, no single employee dominates.

Question 3: What Are the Best- and Worst-Selling Products?

Why this matters: This tells the business what to restock and promote, and what's quietly losing money on the shelf and might need to be discounted, bundled, or dropped.

Best sellers, ranked by quantity sold:

Rank	Product	Qty Sold	Revenue
1	Louisiana Hot Spiced Okra	206,213	3,504,941.00
2	Sir Rodney's Marmalade	205,637	16,653,807.36
3	Teatime Chocolate Biscuits	205,487	1,889,691.42
4	Sirop d'érable	205,005	5,839,809.60
5	Gumbär Gummibärchen	204,761	6,391,018.98

Worst sellers, ranked by quantity sold (lowest first):

Rank	Product	Qty Sold
1	Escargots de Bourgogne	197,673
2	Zaanse koeken	197,889
3	Chef Anton's Cajun Seasoning	198,726
4	Thüringer Rostbratwurst	199,010
5	Röd Kaviar	199,042

Key insight: quantity sold is remarkably similar across almost every product, roughly 197,000 to 206,000 units, regardless of price. This means revenue differences between products are driven mainly by price point, not by how popular a product actually is. Even the lowest performer by quantity still sold close to 200,000 units, so no product stands out as a clear failure worth discontinuing based on this data.

Question 4: Which Shipping Company Is Used Most, and Is It the Fastest?

Why this matters: *Slow delivery damages customer trust and repeat business, so knowing which courier performs best helps the company choose the right shipping partner.*

Shipping Company	Number of Orders	Avg. Days to Ship
United Package	5,475	7.78
Federal Shipping	5,455	7.82
Speedy Express	5,331	7.92

United Package is both the most-used and the fastest shipping company, an ideal alignment. Speedy Express, despite its name, is the slowest of the three and also the least used. This may be worth a closer look to see whether it should be used less, or its service reviewed with the courier.

Question 5: Are There Seasonal Patterns in Order Volume?

Why this matters: *Spotting busy and slow seasons helps the business plan inventory and staffing in advance, instead of scrambling during a rush or sitting on excess stock in slow months.*

Rank	Month	Orders
1	March	1,446
2	August	1,443
3	July	1,412
4	October	1,399
5	December	1,397
6	September	1,389
7	January	1,372
8	May	1,359
9	April	1,331
10	November	1,271
11	June	1,242
12	February	1,221

March, August, and July are the busiest months, while February and June are the slowest. The gap between the busiest and slowest month is relatively small (about 225 orders), so this is a mild seasonal pattern rather than a dramatic swing. The business could plan slightly higher staffing or stock levels around March, July, and August.

Question 6: Do Bigger Discounts Lead to Bigger Order Quantities?

Why this matters: This shows whether discounts are actually working to boost order sizes, or just cutting into profit without changing customer behavior, which helps the business use discounts smarter.

Discount levels with a meaningful number of orders behind them (rare discount values with only 1 to 3 line items were excluded as unreliable):

Discount	Avg. Quantity per Order	Number of Orders
0% (no discount)	25.50	608,445
5%	28.01	185
10%	25.24	173
15%	28.38	157
20%	27.02	161
25%	28.24	154

No. The average order quantity stays roughly the same, between 25 and 28 units, whether there's no discount at all or a discount as high as 25%. There is no clear trend showing that bigger discounts lead to bigger orders. This suggests the discount strategy may not be achieving its intended goal, and it could be worth investigating why discounts are given, or reconsidering the approach.

Conclusion and Skills Demonstrated

This project answered six practical business questions using SQL, and involved catching and correcting two real data quality issues along the way, exactly the kind of careful, verification-driven work a business can rely on.

Skills demonstrated:

- Writing SQL queries involving JOIN, GROUP BY, aggregate functions (SUM, AVG, COUNT), filtering, and sorting
- Working with dates using SQLite's julianday() and strftime() functions
- Identifying and correcting data quality issues before reporting results
- Choosing the right metric for a business question (quantity vs. revenue)
- Translating raw query results into clear, business-relevant recommendations